

Response Letter

Paper ID

482

Paper Title

StrokeGuardian: Hybrid Ensemble ML for Real-Time Brain Stroke Prediction and Clinical Decision Support

Track Name

AI, Computer Networks, Security & IoT

Reviewer-01

Comment 1:

“How was the weighting ratio in the hybrid ensemble (0.4:0.35:0.25) determined—was it empirically optimized or based on performance metrics of individual models?”

Response:

The ratio was determined through an ****empirical optimization process**** guided by performance metrics. Specifically, we allocated an initial weight proportional to the individual out-of-sample performance (e.g., accuracy, F1-score) of each base model on our validation set. We then performed a grid search over a defined parameter space to fine-tune these weights, with the objective of maximizing the ensemble's overall predictive performance. The final ratio of 0.4:0.35:0.25 was found to yield the optimal balance, maximizing the ensemble's performance on the hold-out validation set without inducing overfitting. We acknowledge that this explanation was not sufficiently detailed in the original manuscript.

Action:

The manuscript has been updated to clearly explain the weight selection process in Section IV.C. The description now specifies how the weighting ratios were empirically derived through systematic evaluation.

A detailed explanation has been added:

“We performed an empirical search over multiple candidate weight combinations satisfying

$w_{NN} + w_{SVM} + w_{XGB} = 1$ and evaluated each configuration using 10-fold cross-validation. For every candidate triplet, the mean Accuracy, Recall, and F1-score particularly the Recall of stroke-positive cases were computed. The final ratio of 0.40 : 0.35 : 0.25 was selected because it consistently yielded the best trade-off between

overall accuracy and high sensitivity while maintaining stable performance across validation folds.”

Comment 2:

“Did the study assess the model’s robustness under noisy or incomplete clinical data, which is common in real-world hospital systems?”

Response:

We thank the reviewer for this important observation. The study did not assess model robustness under noisy or incomplete clinical data because the dataset used in this work contained no missing values or measurement noise. We acknowledge that real-world hospital datasets commonly include irregularities such as incomplete records and sensor errors, and this limitation has now been explicitly stated in the manuscript.

Action :

A concise limitations paragraph has been added to the Discussion/Results section to acknowledge the absence of robustness evaluation. Additionally, the Future Work section has been updated to state that future investigations will evaluate the model’s performance under noisy and incomplete clinical conditions using noise-injection and missing-value simulation strategies.

Comment 3:

“The dataset is described as having no missing values, but most real-world clinical datasets contain them—this assumption reduces realism and may overstate performance.”

Response:

We appreciate the reviewer’s observation. The dataset used in this study indeed contains no missing values, which simplifies preprocessing but does not fully represent the complexity of real-world clinical datasets. We acknowledge that actual hospital data typically include missing entries, irregular patterns, and inconsistencies. As a result, the absence of missing-value handling in our dataset may limit the realism of the experimental setup and could lead to performance results that appear higher than what may be expected in real clinical deployment.

Action :

A statement has been added to the Dataset Description and Limitations sections clarifying that the dataset lacks missing values, and this constitutes a limitation of the study. Additionally, the Future Work section has been updated to include plans for incorporating realistic missing-value scenarios and evaluating model performance on datasets that better reflect clinical variability.

Comment 4:

“Include external validation using an independent dataset or cross-hospital testing to confirm the model’s generalizability and real-world reliability.”

Response:

We appreciate the reviewer’s valuable suggestion. The current study evaluates the proposed model using a single dataset and does not include external validation on independent or cross-hospital datasets. We acknowledge that such validation is essential for assessing real-world generalizability and ensuring the model’s reliability across diverse clinical settings.

Action:

The manuscript has been updated to address this limitation. A statement has been added in the Discussion/Limitations section noting the absence of external validation. The Future Work section has also been revised to include plans for validating the model using independent datasets and multi-center (cross-hospital) data to enhance generalizability and real-world applicability.

Reviewer-02**Comment 1:**

“Abstract should be concise”

Response:

The reviewer suggested that the abstract should be concise and clearly present the main contributions. We appreciate this valuable feedback, as it helps improve the clarity and readability of the manuscript.

Action :

Reduced length and removed redundant expressions.Highlighted the core contributions more directly.Improved technical flow and readability. Ensured logical order. Strengthened contribution statement.

Comment 2:

“The main contributions should be clearly stated in the introduction, the section should be richer, and some literature with a critical review should be included in the introduction section.”

Response:

Thank you for the valuable feedback. We agree that the introduction required clearer articulation of the main contributions and a more comprehensive and critical review of the existing literature. We have accordingly revised the section to enrich the background discussion, explicitly highlight the limitations of prior studies, and position the motivation for the proposed work more effectively. Additionally, the contribution points have been clearly listed to provide better clarity on the novelty and significance of the paper.

Action :

The Introduction section has been updated extensively. Specifically:

- A more detailed and critical review of relevant literature has been incorporated to identify existing gaps.
- Clear explanations of the limitations of traditional and machine-learning-based stroke prediction approaches have been added.
- The key contributions of the proposed work have been explicitly listed in bullet-point format for clarity.
- References throughout the section have been corrected and reorganized to match IEEE citation standards.

These modifications improve the richness, clarity, and academic rigor of the Introduction, fully addressing the reviewer’s concerns.

Comment 3:

“The result and analysis section should be re-checked and richer with a critical analysis.”

Response:

We appreciate the reviewer’s insightful comment. We agree that the Results and Analysis section required deeper critical interpretation and clearer explanation of the model behaviors. The earlier version focused mainly on reporting numerical results, which could be expanded to provide stronger analytical insight into error patterns, comparative model weaknesses, and clinical implications.

Action :

To address this comment, the Results and Analysis section has been revised and enriched in the following ways:

1. Expanded critical interpretation

- Detailed explanation added on why Neural Network performs slightly better in accuracy but worse in consistency compared to the ensemble.
- Analysis added on why models like KNN and Naïve Bayes underperform due to data dimensionality and class imbalance effects.

2. Deep error-pattern discussion

- The confusion matrix interpretation has been expanded to highlight borderline-error zones (e.g., glucose and BMI ranges).
- Discussion added on why false negatives are clinically dangerous and how the ensemble minimizes them.

3. Cross-model comparative reasoning

- Additional commentary included on model trade-offs: sensitivity vs. accuracy vs. generalization.
- Explanation added on why SVM and XGBoost support NN in complementary ways inside the ensemble.

4. Clinical relevance of metrics clarified

- Interpretation added on why high Recall is critical for stroke-positive cases.
- ROC–AUC analysis extended to explain real-world discriminative behavior.

5. Stronger comparison with previous literature

- Comparative analysis table extended with a paragraph linking *why* your model outperforms others, not just *how much*.

The revised section now provides richer, deeper, and more critical analysis that fully addresses the reviewer's concerns.

Comment 4

“A section named ‘discussion’ should be added including some previous works related to your work by a comparison table.”

Response:

Thank you for the suggestion. We appreciate the importance of comparing our proposed method with previous studies. We would like to clarify that the manuscript already includes a detailed comparative evaluation in the **Results and Analysis** section. Specifically, Section *Comparative and Error Analysis* presents a comprehensive comparison table (Table~\ref{tab:comparison_previous}) along with critical analysis discussing how the proposed hybrid ensemble model performs relative to prior state-of-the-art approaches.

Since the comparative discussion and interpretation of findings are already integrated into the Results section in a structured and analytical form, we believe this fulfills the intended purpose of a separate Discussion section. For clarity, the comparative explanations in the Results section have been retained and refined where necessary.

Action :

No additional section was created, as the comparison table and discussion already exist under **Results and Analysis**. However, we ensured that this subsection clearly presents:

- performance comparison with previous works,

- critical interpretation, and
- contextual analysis of the proposed model's strengths.

This addresses the reviewer's request while maintaining the logical flow of the manuscript.

Comment 5

"Limitations/challenges of the proposed work could be pointed in the discussion section."

Response:

Thank you for this valuable suggestion. We agree that addressing the limitations of the proposed work is essential for clarity and completeness. We would like to clarify that the manuscript already includes a **dedicated Limitations subsection** within the **Results and Analysis** section. This subsection discusses:

- the absence of noisy or incomplete clinical data testing,
- the assumption of a clean dataset without missing values, and
- the lack of external validation across independent or cross-hospital datasets.

These points explicitly address the practical challenges and generalizability constraints of the proposed framework. Since the limitations are already presented in a focused and structured manner within the Results and Analysis section, we believe this fulfills the reviewer's request without requiring a separate Discussion section.

Action :

No new section was added, as the limitations are already comprehensively discussed in the **Limitations** subsection of the Results and Analysis section. Minor wording refinements were made to ensure the limitations are clearly emphasized and easy for readers to identify.

Comment 6

"Conclusion section should be concise and some future scopes should be included, and the name of the section would be 'Conclusion and Future Scopes'."

Response:

Thank you for the constructive suggestion. We agree that the conclusion should be more concise and clearly include future research directions. We also acknowledge the reviewer's recommendation regarding the section title.

Accordingly, the conclusion has been revised to present a clearer summary of the study's contributions while avoiding redundancies. Additional future scopes have been incorporated to outline the next steps of model enhancement, including noise evaluation, multimodal data integration, external validation, and federated learning. The section title has also been updated to "Conclusion and Future Scopes" to reflect its expanded content.

Action :

- The **Conclusion** section has been rewritten concisely to summarize the main findings.
- A detailed set of **future scopes** has been added, covering robustness evaluation, multimodal extensions, external validation, real-time monitoring, and federated learning.
- The section title has been updated to "**Conclusion and Future Scopes.**"

Comment 7

"References of the whole paper including the reference section should be re-checked."

Response:

Thank you for the remark. We appreciate the importance of maintaining accuracy and consistency in the reference list. The references throughout the manuscript have been carefully reviewed to ensure correctness, proper formatting, and alignment with the citation style.

Comment 8

"All figures should be clearer with a good resolution."

Response:

Thank you for pointing this out. We acknowledge the need for high-quality figures to improve readability. All figures have been reviewed, and high-resolution versions have been prepared to ensure clarity.

Comment 9

“The grammar of the whole manuscript should be rechecked.”

Response:

Thank you for this observation. The manuscript text has been thoroughly reviewed to correct grammatical issues, improve readability, and enhance academic clarity.

Comment 10

“The plagiarism should be checked and it should not be more than 15%.”

Response:

We appreciate this reminder. The manuscript has been checked for plagiarism to ensure originality, and it meets the required similarity threshold.